Documentation for the data pipeline for dashboard:

“SUTV Messaging Campaign”

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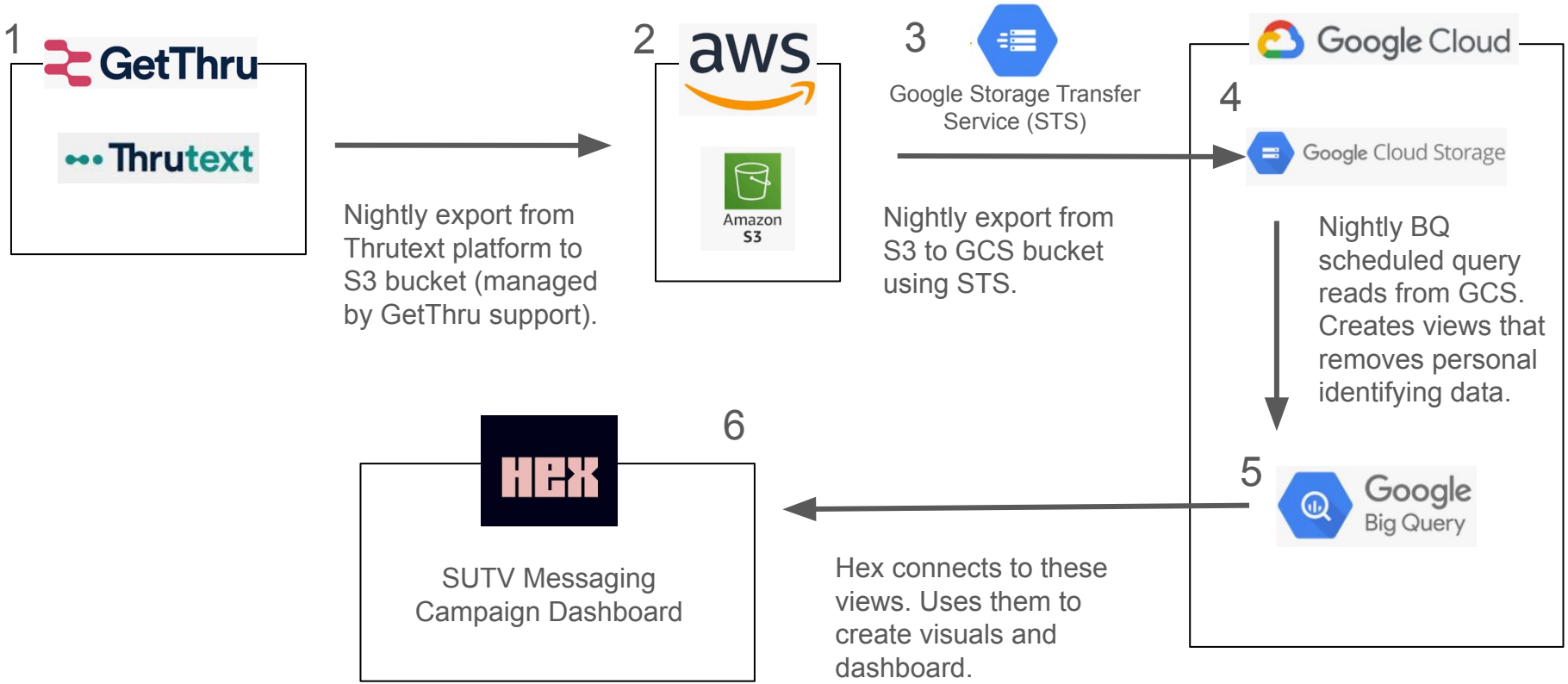
September 2024

Data Pipeline Overview



1. Get Thru

- GetThru, which includes ThruText, supports nightly exports to S3, but not to GCS.
- We worked with GetThru support to set up these exports.
- Please see Google Doc “Shape Up The Vote Mail - Enabling Nightly Exports (S3 vs GCS)” for steps on how to set up an S3 bucket enabled to receive exports.
- Once that is set up, we sent the below email to GetThru support.
- We now get regular, nightly exports. Currently there is no scheduled end date.

Hi GetThru support team,

Per the attached message, I am reaching out to ask for your help in setting up nightly exports between our GetThru account and an S3 bucket that we set up.

Below is the information you require, as stated in the last step of the directions:

1. AWS region: [redacted]
2. S3 bucket: [redacted]
3. Role name: [redacted]
4. GetThru Organization: [redacted]

Would you please let us know if the connection works or if you need further information?

Thanks,

Dennis Elwell

2. AWS: S3 Bucket

- An S3 bucket is AWS' equivalent of Google Cloud Storage (GCS).
- At 8am UTC daily, GetThru sends the following csv exports into this bucket, called "thrutext-exports":
  - caller_activity_details
  - daily_messages
  - daily_surveys
  - opt_outs
- Each csv file represents only data from that particular day.
- S3 was the only option where we could export our ThruText data.
- Since we do not use AWS for any other part of our analytics infrastructure, our next step is to move our data into GCP.

3. Storage Transfer Service

- STS is a Google Cloud Platform service that you can use to transfer files from S3 to GCP.
- This Google Cloud article walks through how we set up the transfer between S3 to GCP: Transfer data from AWS to GCP using Storage Transfer Service
- Since we only needed to set it up once per day, we made it batch-driven, which runs once per day at a scheduled time, rather than event-driven, which runs anytime there is a change in the S3 bucket.
- At 9am UTC daily, we receive exports to GCP. We set this up to end on December 31, 2024.

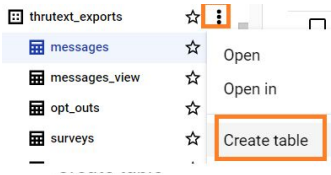
4. Google Cloud Storage

- A GCS bucket is where we store our CSV files transferred from S3.
- The name of the bucket is “thrutext-exports”.
- It is structured exactly the same as the S3 bucket.

5. Big Query (part 1)

- Big Query is GCP’s data warehouse service.
- Within the “shape-up-the-vote” project, we have a dataset called “thrutext_exports”, which contains several tables that are read directly from our GCS bucket’s folders:
  - “messages”
  - “surveys”
- These tables simply read in all the columns and interpret the schema of each of our GCS folders.
- The next page shows an example of how the “messages” table was set up using the BQ console.

How we set up the “messages” and “surveys” tables in BQ console



Source

Create table from
Google Cloud Storage

Select file from GCS bucket or use a URI pattern *
thrutext-exports/daily_messages/*

File format
CSV

Source Data Partitioning

Destination

Project *
shape-up-the-vote

Dataset *
thrutext_exports

Table *
messages

Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes

Table type
External table

Regional / dual region GCS buckets are recommended for External table.

Create a BigLake table using a Cloud Resource connection

Schema

Auto detect

Schema will be automatically generated.

Tags

Tags help you manage and enforce policies on your resources. Tags consist of a unique tag key and a set of tag values. Learn more

SELECT SCOPE

Advanced options

Number of errors allowed
9999

Unknown values

Column name character map
Default

Field delimiter
Comma

Advanced options

Number of errors allowed
9999

Unknown values

Column name character map
Default

Field delimiter
Comma

Quote character
Double quote

Header rows to skip
1

Quoted newlines

Jagged rows

Default Rounding Mode

If not set, the table default rounding mode will be inherited fron
DECIMAL (numeric and bignumeric) columns created. This set
later and if altered, will only apply to columns added after the s

Default Rounding Mode

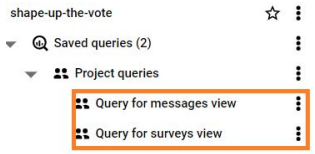
CREATE TABLE

CANCEL

5. Big Query (part 2)

- Once those initial tables were set up, we created “views” tables that did some slight data transformations:
  - They only captured potentially relevant columns and excluded columns that contained personal identifiable information (e.g. people’s names).
  - They took the “address” column and, using regex, broke it down into several separate columns for city, state and zipcode.
- The next pages detail how we did the following:
  - Set up a query for each of the tables.
  - Created a scheduled query that would refresh each of these tables daily at 10am UTC.
  - Note- schedule ends on December 31, 2024.

How we set up saved project queries



We set up two saved queries here.

These are the two saved queries. Note the regex used to extract city, state and zipcode from address.

Also note the “Schedule” button. We click this to set this up on a daily schedule.

Query for messages view

RUN | SAVE QUERY | SHARE | SCHEDULE | OPEN IN | MORE

Classic saved queries will be turned down in March 2025. Use the migration tool to migrate your queries to Bigquery Studio saved queries.

```
SELECT
    inserted_at,
    updated_at,
    message_id,
    message_direction,
    message_error_type,
    billable,
    campaign_id,
    campaign_name,
    conversation_id,
    contact_id,
    shop_id,
    REGEXP_EXTRACT(address, r'([A-Za-z\.]{3}[A-Za-z\.]*),*\s+(?:[A-Za-z]{2})(?:,?\s*\d+(?:[-\s]\d+)?)?\s*$') AS city,
    UPPER(REGEXP_EXTRACT(address, r'\s+([A-Za-z]{2})(?:,?\s*\d+(?:[-\s]\d+)?)?\s*$')) as state,
    REGEXP_EXTRACT(address, r'\s+(?:[A-Za-z]{2},?\s*)(\d{5})(?:[-\s]\d+)?\s*$') as zipcode
FROM
  `thrutext_exports.messages`;
```

Query for surveys view

RUN | SAVE QUERY | SHARE | SCHEDULE | OPEN IN | MORE

Classic saved queries will be turned down in March 2025. Use the migration tool to migrate your queries to Bigquery Studio saved queries.

```
SELECT
  campaign_id,
  campaign_name,
  conversation_id,
  survey_id,
  survey_question,
  survey_type,
  response,
  contact_id,
  updated_at,
  sync_status,
  shop_id,
  REGEXP_EXTRACT(address, r'([A-Za-z\.]{3}[A-Za-z\.]*),*\s+(?:[A-Za-z]{2})(?:,?\s*\d+(?:[-\s]\d+)?)?\s*$') AS city,
  UPPER(REGEXP_EXTRACT(address, r'\s+([A-Za-z]{2})(?:,?\s*\d+(?:[-\s]\d+)?)?\s*$')) as state,
  REGEXP_EXTRACT(address, r'\s+(?:[A-Za-z]{2},?\s*)(\d{5})(?:[-\s]\d+)?\s*$') as zipcode
FROM
  `shape-up-the-vote.thrutext_exports.surveys`
```

How we set up the scheduled queries

SCHEDULE

New scheduled query

Details and schedule

Name for scheduled query *
messages_scheduled_query

Schedule options

Repeat frequency *
Days

At *
10:00

Start now
Start at set time

Start date and run time
9/18/24, 7:26 PM CST

End never
Schedule end time

End date *
12/31/24, 12:00 AM CST

Scheduled to run every day at 10:00 UTC starting now and ending on December 31, 2024 at 12:00:00 AM CST.

Destination for query results

Set a destination table for query results

Dataset *
shape-up-the-vote.thrutext_exports

Table Id *
messages_view

Destination table partitioning field

Destination table write preference

Append to table
Overwrite table

Automatic location selection

Location type

Region
Specify a region to colocate your datasets with other G

Multi-region
Allow BigQuery to select a region within a group to ach

Region

Service account

Service account

Default to use the requesting user crec

Advanced options

Notification options

Send email notifications

Cloud Pub/Sub topic

projects/<project-name>/topics/<topic

SAVE CANCEL

Once saved, you can find them by opening the left-hand side pane and clicking this icon

Scheduled queries

5. Big Query (part 3)

- Once our queries were scheduled, we set them to save as the tables “messages_view” and “surveys_view”.

- These tables are what we connect to in HEX.

6. HEX

- HEX is a data visualization tool that we use to show our “SUTV Message Campaign” dashboard.
- As we create and edit projects, we can connect directly to our tables that are stored in Big Query.
- We can query with SQL and then take the output of those queries and turn them into visualizations.
- The result has been a dashboard, which refreshes as often as our views tables, that our team can use to track our metrics.
- The dashboard is available via this link.

Example of HEX dashboard visuals

SUTV Message Campaign

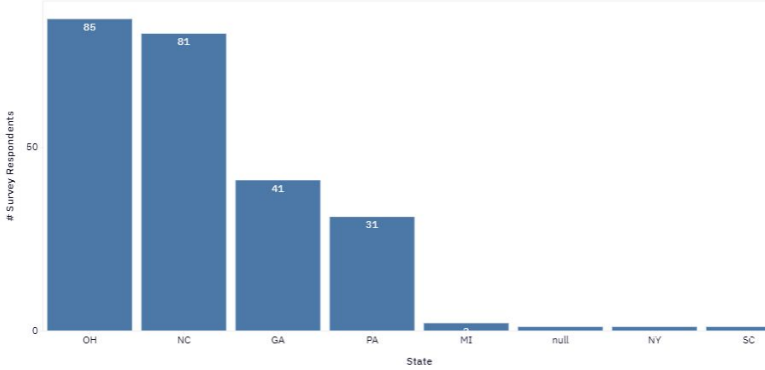
Published {Date} by {Publisher}

Texts | Survey Responses | +

9,009	971	8,223
# Sent Texts	# Response Texts	# Conversations

Sent Texts by Day

243

Survey Respondents

Survey Respondents by State